



**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA**  
**KAKINADA – 533 003, Andhra Pradesh, India**  
**DEPARTMENT OF CIVIL ENGINEERING**

**COURSE STRUCTURE**

**For UG – R20**

**B. TECH - CIVIL ENGINEERING**

*(Applicable for batches admitted from 2020-2021)*



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## COURSE STRUCTURE

### I Year – I SEMESTER

| S. No                | Course Code | Subjects                                            | L           | T | P | Credits |
|----------------------|-------------|-----------------------------------------------------|-------------|---|---|---------|
| 1                    | BSC1101     | Mathematics – I (Calculus & Differential Equations) | 3           | 0 | 0 | 3       |
| 2                    | HSMC1101    | Communicative English                               | 3           | 0 | 0 | 3       |
| 3                    | BSC1102     | Engineering Physics                                 | 3           | 0 | 0 | 3       |
| 4                    | ESC1101     | Engineering Drawing                                 | 1           | 0 | 4 | 3       |
| 5                    | ESC1102     | Engineering Geology (Integrated) (Theory & Lab)     | 2           | 0 | 2 | 3       |
| 6                    | HSMC1102    | English Communication Skills Laboratory             | 0           | 0 | 3 | 1.5     |
| 7                    | BSC1103     | Engineering Physics Lab                             | 0           | 0 | 3 | 1.5     |
| 8                    | ESC1103     | Basics of Civil Engg. Work Shop (Lab)               | 0           | 0 | 3 | 1.5     |
| <b>Total Credits</b> |             |                                                     | <b>19.5</b> |   |   |         |

### I Year – II SEMESTER

| S. No                | Course Code | Subjects                                              | L           | T | P | Credits |
|----------------------|-------------|-------------------------------------------------------|-------------|---|---|---------|
| 1                    | BSC1201     | Mathematics – II (Linear Algebra & Numerical Methods) | 3           | 0 | 0 | 3       |
| 2                    | BSC1202     | Engineering Chemistry                                 | 3           | 0 | 0 | 3       |
| 3                    | ESC1201     | Engineering Mechanics                                 | 3           | 0 | 0 | 3       |
| 4                    | ESC1202     | Programming for Problem Solving Using C               | 3           | 0 | 0 | 3       |
| 5                    | ESC1203     | Building Materials and Concrete Technology            | 3           | 0 | 0 | 3       |
| 6                    | BSC1203     | Engineering Chemistry Lab                             | 0           | 0 | 3 | 1.5     |
| 7                    | ESC1204     | Programming for problem Solving Using C Lab           | 0           | 0 | 3 | 1.5     |
| 8                    | ESC1205     | Building Planning and Computer Aided Building Drawing | 0           | 0 | 3 | 1.5     |
| 9                    | MC1201      | Environmental Science                                 | 2           | 0 | 0 | 0       |
| <b>Total Credits</b> |             |                                                       | <b>19.5</b> |   |   |         |

**\*Breakup of credits for Engineering Graphics/Engineering Workshop shall be 1-0-4 (as per AICTE model curriculum)**

**Universities/Institutions may swap a few courses between 1<sup>st</sup> and 2<sup>nd</sup> semesters to balance the workload of teaching and laboratory schedule.**



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**II Year – I SEMESTER**

| S. No | Course Code | Course Title                                           | L | T | P | Credits     |
|-------|-------------|--------------------------------------------------------|---|---|---|-------------|
| 1     | BSC301      | Mathematics -III (Vector Calculus, Transforms and PDE) | 3 | 0 | 0 | 3           |
| 2     | PCC301      | Strength of Materials - I                              | 3 | 0 | 0 | 3           |
| 3     | PCC302      | Fluid Mechanics                                        | 3 | 0 | 0 | 3           |
| 4     | PCC302      | Surveying and Geometrics                               | 3 | 0 | 0 | 3           |
| 5     | PCC303      | Highway Engineering                                    | 3 | 0 | 0 | 3           |
| 6     | PCC304      | Concrete Technology Lab                                | 0 | 0 | 3 | 1.5         |
| 7     | PCC305      | Highway Engineering Lab                                | 0 | 0 | 3 | 1.5         |
| 8     | PCC306      | Surveying Field Work – I (Lab)                         | 0 | 0 | 3 | 1.5         |
| 9     | SC301       | <b>Skill oriented course*</b>                          | 1 | 0 | 2 | 2           |
| 10    | MC301       | Constitution of India                                  | 2 | 0 | 0 | 0           |
|       |             | <b>Total Credits</b>                                   |   |   |   | <b>21.5</b> |

**II YEAR – II SEMESTER**

| S. No.                                                                               | Course Code | Course Title                                                                                     | L | T | P | Credits     |
|--------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------|---|---|---|-------------|
| 1                                                                                    | PC401       | Complex Variables and Statistical Methods                                                        | 3 | 0 | 0 | 3           |
| 2                                                                                    | PC402       | Strength of Materials -II                                                                        | 3 | 0 | 0 | 3           |
| 3                                                                                    | ES401       | Hydraulics and Hydraulic Machinery                                                               | 3 | 0 | 0 | 3           |
| 4                                                                                    | PC403       | Environmental Engineering                                                                        | 3 | 0 | 0 | 3           |
| 5                                                                                    | PC404       | Managerial Economics & Financial Analysis                                                        | 3 | 0 | 0 | 3           |
| 6                                                                                    | PC405       | Environmental Engineering Lab                                                                    | 0 | 0 | 3 | 1.5         |
| 7                                                                                    | PC406       | Strength of Material Lab                                                                         | 0 | 0 | 3 | 1.5         |
| 8                                                                                    | PC407       | Fluid Mechanics & Hydraulics Machinery Lab                                                       | 0 | 0 | 3 | 1.5         |
| 9                                                                                    | SC401       | <b>Skill oriented course*</b>                                                                    | 1 | 0 | 2 | 2           |
| 10                                                                                   | PR401       | Industrial/Research Internship (Mandatory) 2 Months...<br>to be evaluated in III year I semester |   |   |   |             |
|                                                                                      |             | <b>Total Credits</b>                                                                             |   |   |   | <b>21.5</b> |
| <b>Honors/ Minor courses<br/>(The hours distribution can be 3-0-2 or 3-1-0 also)</b> |             |                                                                                                  | 3 | 1 | 0 | 4           |



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**III YEAR – I SEMESTER**

| S. No                                                                                      | Course Code | Course Title                                                                                | L | T | P | Credits     |
|--------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------|---|---|---|-------------|
| 1                                                                                          | PC501       | Structural Analysis                                                                         | 3 | 0 | 0 | 3           |
| 2                                                                                          | PC502       | Design and Drawing of Reinforced Concrete Structures                                        | 3 | 0 | 0 | 3           |
| 3                                                                                          | PC503       | Geotechnical Engineering-1                                                                  | 3 | 0 | 0 | 3           |
| 4                                                                                          | OE501       | Open Elective Course/Job Oriented Elective (OE-1)                                           | 3 | 0 | 0 | 3           |
| 5                                                                                          | PE501       | Professional Elective course - I                                                            | 3 | 0 | 0 | 3           |
| 6                                                                                          | PC504       | Professional Core courses Lab Survey Camp (Field work)                                      | 0 | 0 | 3 | 1.5         |
| 7                                                                                          | PC505       | Geotechnical Engineering Lab                                                                | 0 | 0 | 3 | 1.5         |
| 8                                                                                          | PC501       | Skill advanced course: Design of Special Structure, Chimney, Hinge Tanks, spill ways etc.,  | 1 | 0 | 2 | 2           |
| 9                                                                                          | MC501       | Essence of Indian Traditional Knowledge                                                     | 2 | 0 | 0 | 0           |
| 10                                                                                         | PR501       | Summer Internship 2Months (Mandatory) after second year (to be evaluated during V semester) |   |   |   | 1.5         |
|                                                                                            |             | Total Credits                                                                               |   |   |   | <b>21.5</b> |
| <b>Honors/ Minor courses</b><br><b>(The hours distribution can be 3-0-2 or 3-1-0 also)</b> |             |                                                                                             | 3 | 1 | 0 | 4           |



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**III YEAR – II SEMESTER**

| S. No                                                                               | Course Code | Course Title                                                                                 | L | T | P | Credits     |
|-------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------|---|---|---|-------------|
| 1                                                                                   | PC601       | Design and Drawing of Steel Structures                                                       | 3 | 0 | 0 | 3           |
| 2                                                                                   | PC602       | Water Resource Engineering                                                                   | 3 | 0 | 0 | 3           |
| 3                                                                                   | PC603       | Geotechnical Engineering-2                                                                   | 3 | 0 | 0 | 3           |
| 4                                                                                   | PE601       | Professional Elective course -II                                                             | 3 | 0 | 0 | 3           |
| 5                                                                                   | OE601       | Open Elective Course/Job oriented Elective (OE-2)                                            | 3 | 0 | 0 | 3           |
| 6                                                                                   | PC604       | Professional Core courses Lab (Estimation, Costing and Contracts)                            | 0 | 0 | 3 | 1.5         |
| 7                                                                                   | PC605       | Professional Core courses Lab (Remote Sensing & GIS Lab)                                     | 0 | 0 | 3 | 1.5         |
| 8                                                                                   | PC606       | Professional Core courses Lab Civil Engineering Practice                                     | 0 | 0 | 3 | 1.5         |
| 9                                                                                   | SC601       | Skill advanced course/ soft skill course: Computational Tools                                | 1 | 0 | 2 | 2           |
| 10                                                                                  | MC601       | Employability Skills                                                                         | 2 | 0 | 0 | 0           |
| 11                                                                                  | PR601       | Industrial/Research Internship (Mandatory) 2 Months... to be evaluated in IV year I semester |   |   |   |             |
|                                                                                     |             | Total Credits                                                                                |   |   |   | <b>21.5</b> |
| <b>Honors/ Minor courses<br/>(The hours distribution can be 3-0-2 or 3-1-0also)</b> |             |                                                                                              | 3 | 1 | 0 | 4           |



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**IV YEAR – I SEMESTER**

| S. No                                                                                   | Course Code | Course Title                                                                                               | L | T | P | Credits   |
|-----------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------|---|---|---|-----------|
| 1                                                                                       | PE701       | Professional Core course - III                                                                             | 3 | 0 | 0 | 3         |
| 2                                                                                       | PE702       | Professional Core course - IV                                                                              | 3 | 0 | 0 | 3         |
| 3                                                                                       | PE703       | Professional Core course - V                                                                               | 3 | 0 | 0 | 3         |
| 4                                                                                       | OE701       | Open Elective Courses/ Job oriented elective (OE-III)                                                      | 2 | 0 | 2 | 3         |
| 5                                                                                       | OE702       | Open Elective Course/Job oriented elective (OE-IV)                                                         | 2 | 0 | 2 | 3         |
| 6                                                                                       | HSC701      | Universal Human Values-2: Understanding Harmony                                                            | 3 | 0 | 0 | 3         |
| 7                                                                                       | SC701       | Skill advanced course/ soft skill course:<br>Project planning & town planning,                             | 1 | 0 | 2 | 2         |
| 8                                                                                       | PR701       | Industrial/Research Internship 2 Months (Mandatory) after third year (to be evaluated during VII semester) |   |   |   | 3         |
|                                                                                         |             | Total Credits                                                                                              |   |   |   | <b>23</b> |
| <b>Honors/ Minor courses</b><br><b>(The hours distribution can be 3-0-2or3-1-0also)</b> |             |                                                                                                            | 3 | 1 | 0 | 4         |

**IV YEAR – II SEMESTER**

| S. NO | CATEGORY      | COURSE TITLE          | L | T | P/D | C         |
|-------|---------------|-----------------------|---|---|-----|-----------|
| 1     | Major Project | PROJ                  | - | - | -   | 12        |
|       |               | INTERNSHIP (6 Months) |   |   |     |           |
|       |               | Total Credits         |   |   |     | <b>12</b> |

**Major Project-12;**

**Internship-6 Months;**

**Total-12**



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**Professional Electives R20 (5 PE x 3 = 15 Credits)**

(Department can offer Maximum 2 Subjects from Each PE, elected by the students)

**Note: Student must choose subjects which were not opted earlier  
 PE starts from III-I**

| <b>Professional Elective-I</b>          | <b>Professional Elective-II</b>   | <b>Professional Elective-III</b>   | <b>Professional Elective-IV</b>        | <b>Professional Elective-V</b>                     |
|-----------------------------------------|-----------------------------------|------------------------------------|----------------------------------------|----------------------------------------------------|
| a) Construction Technology & Management | a) Advanced Structural Analysis   | a) Advanced Structural Engineering | a) Ground Improvement Techniques       | a) Design & Drawing of Irrigation Structures       |
| b) Remote Sensing and GIS               | b) Architecture and Town Planning | b) Bridge Engineering              | b) Geo-Spatial Technologies            | b) Earth & Rock fill Dams                          |
| c) Environmental Impact Assessment      | c) Road Safety Engineering        | c) Structural Dynamics             | c) Disaster Management & Mitigation    | c) Urban Hydrology                                 |
| d) Low Cost Housing                     | d) Traffic Engineering            | d) Urban Transportation Planning   | d) Soil Dynamics & Machine Foundations | SWAYAM / NPTEL / MOOCS COURSES (12 weeks duration) |



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**HONORS R20 (Starts from II-II)**

**(4x4+2 MOOCS/NPTELx2=20 Credits) for Civil Engg. Students**

**Note: Student must choose subjects which were not opted earlier**

**(Any FOUR courses may be chosen by the Student from each Pool)**

| <b>Structural Engineering</b>       | <b>Geotechnical Engineering</b> | <b>Environment and Water Resource Engineering</b> | <b>Transportation Engineering</b>       | <b>Construction Technology and Management</b> |
|-------------------------------------|---------------------------------|---------------------------------------------------|-----------------------------------------|-----------------------------------------------|
| Finite Element Methods              | Reinforced Soil Structures      | Urban Hydrology                                   | Traffic Engineering                     | Construction Technology and Management        |
| Matrix Analysis of Structures       | Advanced Foundation Engineering | Water and Wastewater Management                   | Intelligent Transportation System       | Architecture & Town Planning                  |
| Earthquake Resistant Design         | Earth Retaining Structures      | Water Resources Planning and Management           | Railway, Harbor and Airport Engineering | Repairs and Maintenance of Structures         |
| Pre-stressed concrete               | Geoenvironmental Engineering    | Environmental Impact Assessment                   | Pavement Management System              | Disaster Management and Mitigation            |
| Repair & Retro-fitting of Buildings | Earth & Rock Fill Dams          | Air Pollution and Control                         | Urban Transportation Planning           | Precast and Prefabricated Structures          |





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**OPEN ELECTIVES R20**  
**(4 OE x 3 = 12 Credits)**

**Note: Student must choose subjects which were not opted earlier.**  
**(OE Starts from III-I)**

**Open Elective-1 in III year I semester:**

- a) Strength of Materials
- b) Fluid Mechanics
- c) Surveying and Geomatics

**Open Elective-2 in III year II semester:**

- a) Elements of Civil Engineering
- b) Environmental Engineering
- c) Disaster Management

**Open Elective-3 in IV year I semester:**

- a) Highway Engineering
- b) Safety Engineering
- c) Environmental Management
- d) Urban Planning

**Open Elective-4 in IV year II semester:**

- a) Water Resource Engineering
- b) Hydraulics and Hydraulic Machinery
- c) Green Technologies
- d) Remote Sensing & GIS



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**Minor R20 (Starts from II-II)**  
**(4x4+2 MOOCS/NPTELx2 = 20 Credits)**

**Note: Student must choose subjects which were not opted earlier**

**Minor-1 in II year II semester:**

- a) Construction Technology and Infrastructure Management
- b) Seismology and Earthquake Engineering

**Minor-2 in III year I semester:**

- a) Environmental Engineering and Management
- b) Solid Mechanics

**Minor-3 in III year II semester:**

- a) Railways, Harbours and Docks
- b) Architecture and Smart City

**Minor-4 in IV year I semester:**

- a) Irrigation Engineering
- b) Geoinformatics